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Editor-in-Chief
Knowledge-Based Systems
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Dear Editor-in-Chief,

I am pleased to submit my manuscript entitled “FDR-Controlled Knowledge Graph Completion: Statistical Inference for Reliable Link Prediction” for consideration for publication in Knowledge-Based Systems.

Innovation and Significance. This work introduces a false discovery rate (FDR) control framework that reformulates knowledge graph (KG) link prediction evaluation as a large-scale multiple hypothesis testing problem. While KG embedding models have been studied for over a decade, no existing evaluation protocol quantifies which individual predictions are statistically distinguishable from random noise—nor what fraction of a model’s output carries genuine knowledge signal. We fill this gap by constructing empirical p -values from embedding scores via permutation-based negative sampling, then applying Benjamini-Hochberg FDR control, Storey’s q -value estimation, and Efron’s empirical Bayes local FDR. The resulting three-level evaluation protocol produces per-triple, per-relation, and per-model reliability profiles that complement—rather than replace—existing ranking metrics. The framework is compatible with major embedding paradigms and computationally accessible: the full pipeline completes in 3–5 minutes per model on consumer hardware.

Fit with Knowledge-Based Systems. This manuscript directly aligns with KBS’s focus on knowledge-driven methodologies. The framework addresses a core problem in knowledge engineering: determining which KG completion predictions can be trusted as reliable knowledge for downstream knowledge-based systems. The FDR-based evaluation protocol is not merely an alternative metric but a diagnostic instrument that reveals what fraction of a KG embedding model’s predictions carry genuine knowledge signal—information critical for applications such as clinical decision support, drug repurposing, and financial risk monitoring. The work bridges the gap between statistical methodology and knowledge representation, contributing to the principled construction and evaluation of knowledge-based systems. Furthermore, the approach is grounded in the KBS community’s own calls for more rigorous KG evaluation (Jarnac et al., 2025; Biswas et al., 2023).

Key Findings. Experiments on WN18RR and FB15k-237 across four models (TransE, RotatE, ComplEx, ConvE) reveal that: (1) the estimated null proportion $\hat{\pi}_0$ ranges from 0.259 (RotatE, WN18RR) to 0.851 (ComplEx, FB15k-237), indicating that 15–85% of test triples are statistically indistinguishable from noise, (2) per-relation FDR rejection rates span 9.1%–96.5% under a single model, exposing extreme heterogeneity invisible to MRR, and (3) MRR and FDR produce conflicting model rankings—ConvE trails TransE on MRR (0.469 vs. 0.560) yet achieves $2.8\times$ the BH rejection rate. These findings are validated through sensitivity analyses ($K \in \{25, 50, 100\}$, $\lambda \in \{0.3, 0.5, 0.7\}$) and a diagnostic decomposition of the FDR component stack (BH, Storey q -values, Efron local FDR).

Declarations.

- This manuscript has not been published previously and is not under consideration for publication elsewhere. All authors have approved its submission.
- The author declares no competing financial interests or personal relationships that could have influenced this work.
- A Data Availability Statement and CRediT Author Statement are included with the submission.
- The complete source code, trained model checkpoints, and per-relation FDR statistics are publicly available at <https://github.com/Jackxiaozhiren/fdr-kg>. All experiments are reproducible with

random seed 42.

- Suggested reviewers are listed in the accompanying `referee_suggestions.txt`.

I believe this work will be of significant interest to the KBS readership, particularly researchers working on knowledge graph construction, evaluation, and reliability assessment for knowledge-based applications. The framework is designed as a practical tool that researchers can adopt immediately, and we have released it as open-source software to facilitate community adoption.

Thank you for your time and consideration.

Sincerely,

Zhiren Xiao